

Program No 11

Aim

To implement a concurrent Time Server application using UDP in Python, where the client sends a time request to the server and the server sends its current system time back to the client.

Algorithm

Server

1. Start the program.
2. Import the socket and datetime modules.
3. Create a UDP socket using SOCK_DGRAM.
4. Bind the server socket to IP address 127.0.0.1 and port 5000.
5. Wait for a request from the client using recvfrom().
6. Get the current system time using datetime.now().
7. Convert the time into a string format.
8. Send the time to the client using sendto().
9. Repeat the process to handle further requests.

Client

1. Start the program.
2. Import the socket module.
3. Create a UDP socket.
4. Specify the server address as 127.0.0.1 and port 5000.
5. Send a "TIME" request to the server using sendto().
6. Receive the time from the server using recvfrom().
7. Display the received server time.
8. Close the socket.
9. Stop.

Server program

```
import socket
from datetime import datetime
server = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server.bind(("127.0.0.1", 5000))
print("Server is running...")
while True:
    data, address = server.recvfrom(1024)
    print("Request received from client")
    time = datetime.now().strftime("%H:%M:%S")
    server.sendto(time.encode(), address)
```

Client Program

```
import socket
client = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
```

```
server_address = ("127.0.0.1", 5000)
client.sendto(b"TIME", server_address)
data, address = client.recvfrom(1024)
print("Server Time:", data.decode())
client.close()
```